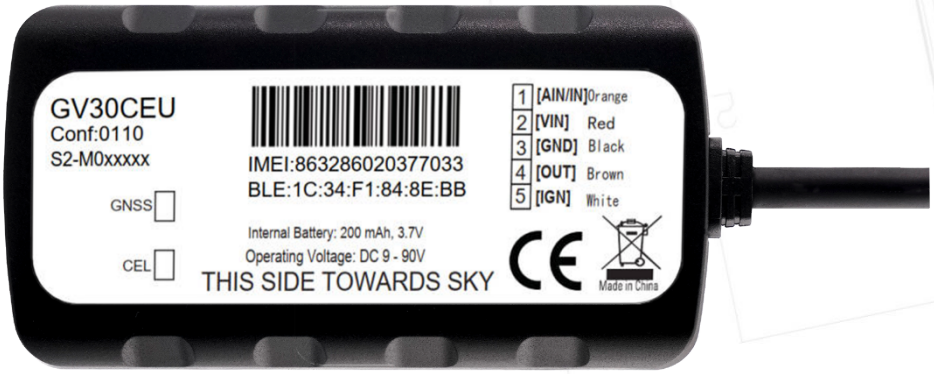
GV30CEU User Manual

GSM/GPRS/LTE Cat1/GNSS Tracker

TRACGV30CEUUM001

Version: 1.00



Driving Smarter IoT

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0. Revision History

Revision	Date	Author	Description of Change
1.00	2025-06-25	Nancy Yin	Initial

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1. Introduction

The GV30CEU is a compact GNSS vehicle tracking device that supports EGPRS and LTE Cat 1. It is designed for a wide variety of applications such as stolen vehicle recovery, motorcycle monitoring and other basic tracking applications. The built-in GNSS receiver has superior sensitivity and fast initial positioning. The full-featured @Track Air Interface Protocol provides the complete documentation, so it's easy to learn system integration. The protocol supports a wide variety of reports including emergency alarm, geo-fence boundary crossings, external power supply monitoring and position reports.

1.1. Reference

Table 1: GV30CEU Protocol Reference

SN	Document Name	Remark
[1]	GV30CEU @Track Air Interface Protocol	The air protocol interface between GV30CEU and the backend server.

1.2. Terms and Abbreviations

Table 2: Terms and Abbreviations

Abbreviation	Description
AIN/IN1	Analog input/Digital input1
PWR	External DC power input
GND	Power and digital ground
OUT	Digital output
IGN	Ignition input, positive trigger

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2. Product Overview

2.1. Appearance

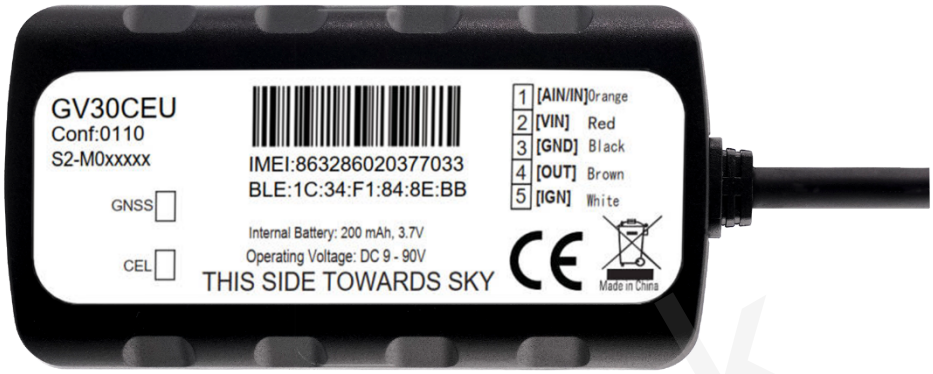


Figure 1: Appearance of GV30CEU

2.2. Interface Definition

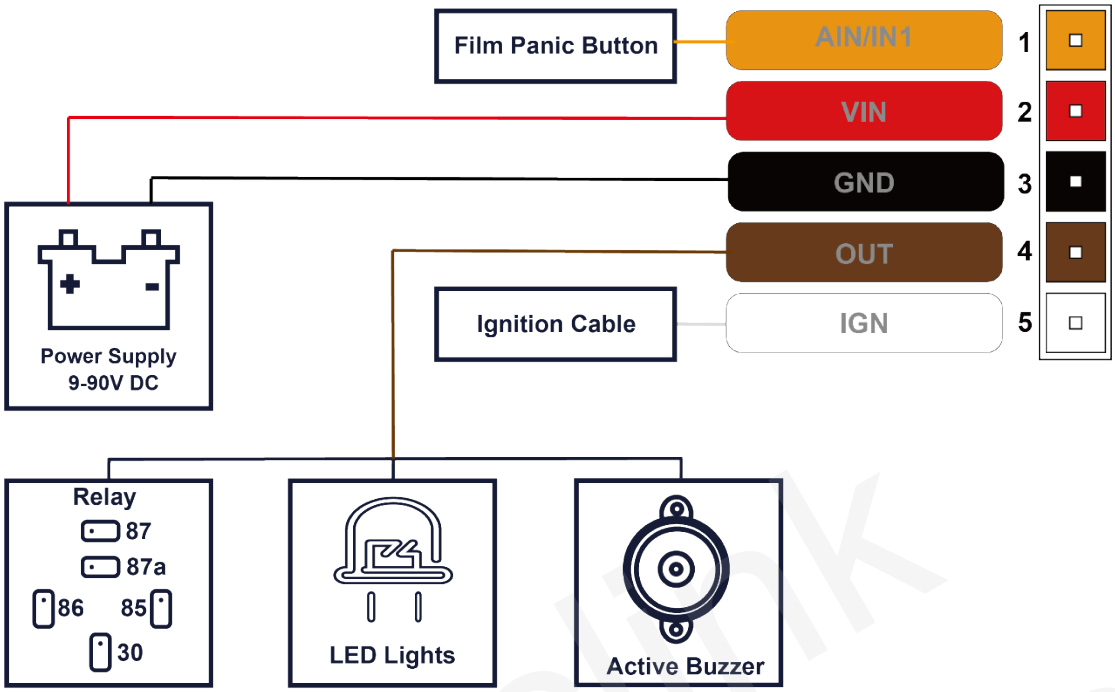
GV30CEU has a 5-PIN interface connector. It contains the connections for power and I/O. The sequence and definition of the 5-PIN connector is shown in the following figure:

Table 3: Description of 5-PIN Connections

Interface	PIN Number	PIN Name	Description
	1	AIN/IN1	Analog Input/Digital Input1, negative trigger
	2	VIN	External DC Power Input, 9-90V
	3	GND	Power and digital ground
	4	OUT	Digital Output, open drain, 150mA max
	5	IGN	Ignition Input, positive trigger

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2.3. Wiring Scheme



2.4. LED Description

GV30CEU has two status LED lights, which are GNSS LED and CEL LED.

Table 4: LED Description

LED	Device Status	LED Status
GNSS	GNSS chip is powered off.	OFF
	GNSS sends no data or data format error occurs.	Slow flashing
	GNSS chip is searching GNSS information.	Fast flashing
	GNSS chip has gotten GNSS information.	ON
CEL	The device is searching network.	Fast flashing
	The device has been registered on the network.	Slow flashing
	SIM card needs pin code to unlock.	ON

Note:

1. The two LED lights can be configured to be turned off after a period time by using the configuration tool.
2. Fast flashing: About 100ms ON/ 200ms OFF.
3. Slow flashing: About 200ms ON/ 1000ms OFF.

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2.5. Power Connection

VIN(Red)/GND(Black) are the power input pins. The input voltage range for this device is from 9V to 90V. The device is designed to be installed in vehicles that operate on 9V to 90V vehicle without the need for external transformers.

2.6. Ignition Detection

Table 5: Electrical Characteristics of Ignition Detection

Logical State	Electrical State
Active	9V to 90V
Inactive	0V to 3V or Open

IGN(White) is used for ignition detection. It is strongly recommended to connect this pin to ignition key RUN position.
An alternative to connecting to the ignition switch is to find a non-permanent power source that is only available when the vehicle is running, for example, the power source for the FM radio.
IGN signal can be configured to start transmitting information to the backend server when the ignition is on, and enter power saving mode when the ignition is off.

2.7. Digital Output

The digital output is an open-drain digital output. The maximum drain current for the device is 150mA.

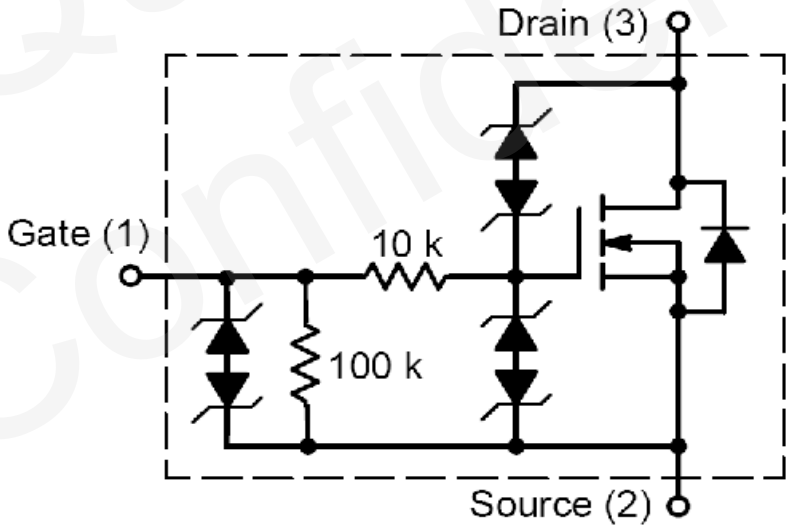


Figure 2: Digital Output Internal Drive Circuit

Table 6: Electrical Characteristics of Digital Output

Logical State	Electrical Characteristics
Enable	<1.5V @150mA
Disable	Open drain

2.8. Analog Input/Digital Input

There is one input can be configured as an Analog input or a digital input on GV30CEU.

F or the digital input, it is a negative trigger.

For the Analog input, the range of input voltage is from 0V to 16V.

Table 7: Electrical Characteristics of Digital Input

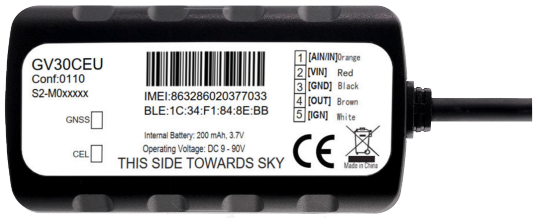
Logical State	Electrical Characteristics
Active	0V to 0.5V
Inactive	Open

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3. Get Started

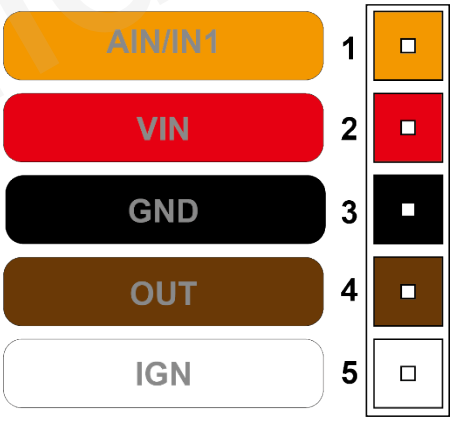
3.1. Parts List

Table 8: Parts List

Name	Picture
GV30CEU Locator	71.5mm*39mm*16.2mm 
MICRO USB CABLE (Optional)	

3.2. External Cable Interface

Table 9: User Cable Color Definition

Definition	Color	PIN No.	Cable
AIN/IN1	Orange	1	
VIN	Red	2	
GND	Black	3	
OUT	Brown	4	
IGN	White	5	

3.3. Turn on/Turn off

- Turn on: Connect the device to external power, and it will be turned on automatically. CEL LED light will be on.

- Turn off: Set <*Battery Working Mode*> in **AT+GTCFG** to **0** to disable the backup battery, and then disconnect the device from the external power.

3.4. Open the Case



Figure 3: Open the Case

Insert the triangular-pry-opener into the gap of the case as shown in the figure, and push the opener up until the case is unsnapped.

When GV30CEU is shipped, the case of GV30CEU is not closed completely, which allows the user to open it relatively easily for SIM card and battery installation.

3.5. Install a SIM Card



Figure 4: SIM Card Installation

Open the case and ensure the unit is not powered. Slide the holder up to open the SIM card holder. Insert the SIM card into the holder as shown above with the gold-colored contact area facing down. Take care to align the cut mark. Close the SIM card holder. Close the case.

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3.6. Install the Internal Battery



Figure 5: Battery Installation

3.7. Close the Case



Figure 6: Close the Case

Put the upper cover on the lower cover, and press the covers to make sure they are closed completely.

3.8. Motion Sensor Direction

GV30CEU has an internal 3-axis accelerometer supporting driving behavior monitoring, crash detection and motion detection. The following figure shows the directions of the motion sensor.

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Figure 7: Motion Sensor Direction

Note:

1. The opposite direction of the cable bundle is the positive direction of the X-axis.
2. The Z-axis is in the positive direction above the front housing surface.
3. The positive directions of the three axes are perpendicular to each other, as shown in the figure above.

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4. Troubleshooting and Safety Information

4.1. Troubleshooting

Table 10: Solutions to Possible Trouble

Trouble	Possible Reason	Solution
After GV30CEU is turned on, the CEL LED always flashes quickly.	The signal is too weak, and GV30CEU cannot be registered on the network.	Please move GV30CEU to places with good GSM coverage.
Messages cannot be reported to the backend server.	The IP address or port of the backend server is wrong.	Make sure the IP address for the backend server is an identified address on the Internet.
Unable to power off GV30CEU.	Unable to power off GV30CEU if charger is connected.	Disconnect charger, and try again.
GV30CEU cannot get successful GNSS fix.	The GNSS signal is weak.	Please move GV30CEU to a place with open sky. It is better to let the top surface (the surface with LED indicator) face the sky.

4.2. Safety Information

- Please do not disassemble the device by yourself.
- Please do not put the device on overheated or too humid place, and avoid exposure to direct sunlight. Too high temperature will damage the device or even cause battery explosion.
- Please do not use GV30CEU on the airplane or near medical equipment.

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